

Preface

Generative AI has fundamentally changed how software developers work. Large Language Models (LLMs) like GPT-4, Claude, and Gemini are no longer curiosities; they are practical tools that developers use daily for code generation, debugging, documentation, and building intelligent applications.

This handbook distills a comprehensive 4-day training curriculum into a self-paced reference that any software developer can pick up and immediately apply. Every chapter combines conceptual depth with hands-on code you can run against real LLM APIs.

Who This Book Is For

- Software developers who want to build applications powered by LLMs
- Engineers evaluating prompt engineering, RAG, and chatbot architectures
- Technical leads who need to understand production concerns: cost, security, and ethics
- Anyone with Python experience who wants a practical, code-first introduction to Generative AI

What You Will Learn

- How LLMs work at a level sufficient to use them effectively (no PhD required)
- Prompt engineering techniques that produce reliable, structured output
- Building conversational AI and RAG-powered applications from scratch
- Production best practices: security, cost optimization, bias mitigation, and evaluation

Prerequisites

- Working knowledge of Python (functions, classes, packages)
- A terminal and code editor (VS Code recommended)

- An API key from at least one LLM provider (OpenAI, Google Gemini, or Anthropic Claude)

Conventions Used in This Book

- `code blocks` denote Python code, API parameters, or terminal commands
- **Bold** highlights key terms on first introduction
- Diagrams use Mermaid notation rendered as SVG